

JEE MAIN REVISION MASTER: INVERSE TRIGONOMETRIC FUNCTIONS

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 22 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Inverse Trigonometric Functions — Core Principles & Governing Laws	15 PYQs	Core Principle
High	Inverse Trigonometric Functions — Standard Calculations & Problem Archetypes	8 PYQs	Quantitative Analysis
Standard	Inverse Trigonometric Functions — Applied Cases, Reactions & Traps	0 PYQs	Applied Topics

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Inverse Trigonometric Functions — Core Principles & Governing Laws	$f(x)=\cos^{-1}\sqrt{x-1}$, $\quad f'(\sqrt{3})$, $\quad \sec^{-1}x$
Inverse Trigonometric Functions — Standard Calculations & Problem Archetypes	$\sin^{-1}x$, $\quad \cos(\sin^{-1}\frac{\pi}{3}+\cos^{-1}\frac{\pi}{3})$, $\quad \cot^{-1}\frac{\sqrt{1-\sin x}+\sqrt{1+\sin x}}{\sqrt{1-\sin x}-\sqrt{1+\sin x}}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Inverse Trigonometric Functions — Core Principles & Governing Laws (Very High Yield — 15 PYQs)

Core Principle: Fundamental definitions, core postulates, and governing physical/chemical relations in Inverse Trigonometric Functions.

■ **Caution / JEE Pitfall:** Carefully verify units, standard reference states, and sign conventions in Inverse Trigonometric Functions.

■ **Speed Trick:** Focus on high-yield formulas and standard limiting approximations tested in Inverse Trigonometric Functions PYQs.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): " $\tan^{-1}((1/1+1 \cdot 2)) + \tan^{-1}((1/1+2 \cdot 3)) + \dots + \tan^{-1}((1/1+n(n+1))) = \dots$ " [Ans: A]
- [2026 Shift] Q#2 (MCQ): "If $\sin^{-1}x + \sin^{-1}y = (\pi/2)$, then x^2 is equal to..." [Ans: A]
- [2025 Shift] Q#3 (MCQ): " $\sec^2(\tan^{-1}2) + \operatorname{cosec}^2(\cot^{-1}3) = \dots$ " [Ans: N/A]

3.2 Inverse Trigonometric Functions — Standard Calculations & Problem Archetypes (High Yield — 8 PYQs)

Core Principle: Quantitative numerical problem-solving techniques and standard question patterns frequently seen in Inverse Trigonometric Functions.

■ **Caution / JEE Pitfall:** Watch for algebraic rounding, stoichiometric balancing, and boundary condition checks.

■ **Speed Trick:** Dimensional consistency checks and ratio comparisons can quickly eliminate incorrect options.

Representative Source PYQs:

- [2025 Shift] Q#4 (MCQ): " $2\cos^{-1}x = \sin^{-1}(2x\sqrt{1-x^2})$ is valid for all values of ' x ' satisfying..." [Ans: N/A]
- [2024 Shift] Q#2 (MCQ): "If $\cos^{-1}x + \cos^{-1}y + \cos^{-1}z = 3\pi$, then $x(y+z) + y(z+x) + z(x+y)$ equals to..." [Ans: N/A]
- [2023 Shift] Q#1 (MCQ): "The value of $\cot^{-1}\left[\frac{\sqrt{1-\sin x} + \sqrt{1+\sin x}}{\sqrt{1-\sin x} - \sqrt{1+\sin x}}\right]$, where $x \in (0, (\pi/4))$ is..." [Ans: N/A]

3.3 Inverse Trigonometric Functions — Applied Cases, Reactions & Traps (Standard Yield — 0 PYQs)

Core Principle: Special cases, experimental observations, and conceptual traps designed to test deep understanding of Inverse Trigonometric Functions.

■ **Caution / JEE Pitfall:** Read questions carefully for keywords like 'incorrect', 'not true', or non-standard conditions.

■ **Speed Trick:** Elimination of extreme or unphysical answer choices improves accuracy in assertion-reason and multiple-statement questions.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): " $\tan^{-1}((1/1+1 \cdot 2)) + \tan^{-1}((1/1+2 \cdot 3)) + \dots + \tan^{-1}((1/1+n(n+1))) = \dots$ " [Ans: A]
- [2026 Shift] Q#2 (MCQ): "If $\sin^{-1}x + \sin^{-1}y = (\pi/2)$, then x^2 is equal to..." [Ans: A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

$$\tan^{-1}\left(\frac{1}{1+1 \cdot 2}\right) + \tan^{-1}\left(\frac{1}{1+2 \cdot 3}\right) + \dots + \tan^{-1}\left(\frac{1}{1+n(n+1)}\right) =$$

- (A) $\tan^{-1}\left(\frac{n}{n+2}\right)$ [Correct]
- (B) $\tan^{-1}\left(\frac{n+1}{n}\right)$
- (C) $\tan^{-1}\left(\frac{n}{n+1}\right)$
- (D) $\tan^{-1}\left(\frac{n+2}{n}\right)$

Correct Answer: A

Solution Analysis: Given sum is $\tan^{-1}\left(\frac{1}{1+1 \cdot 2}\right) + \tan^{-1}\left(\frac{1}{1+2 \cdot 3}\right) + \dots + \tan^{-1}\left(\frac{1}{1+n(n+1)}\right)$. The general term is $\tan^{-1}\left(\frac{1}{1+k(k+1)}\right)$. Using the identity $\tan^{-1}x - \tan^{-1}y = \tan^{-1}\left(\frac{x-y}{1+xy}\right)$, take $x=k+1$ and $y=k$. Then, $\tan^{-1}(k+1) - \tan^{-1}k = \tan^{-1}\left(\frac{(k+1)-k}{1+k(k+1)}\right) = \tan^{-1}\left(\frac{1}{1+k(k+1)}\right)$. So each term can be written as $\tan^{-1}\left(\frac{1}{1+k(k+1)}\right) = \tan^{-1}(k+1) - \tan^{-1}k$. Now the sum becomes $[\tan^{-1}2 - \tan^{-1}1] + [\tan^{-1}3 - \tan^{-1}2] + \dots + [\tan^{-1}(n+1) - \tan^{-1}n]$. All middle terms cancel, leaving $\tan^{-1}(n+1) - \tan^{-1}1$.

Benchmark Problem 2: KCET 2025 (Q#4)

$2\cos^{-1}x = \sin^{-1}(2x\sqrt{1-x^2})$ is valid for all values of 'x' satisfying

- (A) $0 \leq x \leq \frac{1}{\sqrt{2}}$
- (B) $-1 \leq x \leq 1$
- (C) $0 \leq x \leq 1$
- (D) $\frac{1}{\sqrt{2}} \leq x \leq 1$

Correct Answer: None

Solution Analysis: To determine the values of 'x' that satisfy the equation $2\cos^{-1}x = \sin^{-1}(2x\sqrt{1-x^2})$, let's analyze the components: Let $x = \cos\theta$. This implies $\theta = \cos^{-1}x$. The equation can be expressed as $\sin^{-1}(2\cos\theta \sin\theta)$, which simplifies to $\sin^{-1}(\sin 2\theta)$. Hence, $2\theta = \sin^{-1}(\sin 2\theta)$. For $\sin^{-1}(\sin 2\theta)$ to equal $2\cos^{-1}x$, i.e., 2θ , 2θ must lie within the range $[0, \pi]$. Given: $\theta = \cos^{-1}x$, the valid range for θ is $[0, \pi/4]$. Therefore, $\cos^{-1}x \in [0, \pi/4]$. This implies: x lies within $[\frac{1}{\sqrt{2}}, 1]$.